ROCSCIENCE

Case3_with_surcharge_50kPa
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:34:42 am
Software Version: 9.041

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Slide2 Analysis Information

Case3_with_surcharge_50kPa

Project Summary

File Name:	Case3_with_surcharge_50kPa.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:04.726s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:34:42 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	25
Tolerance:	0.0001
Maximum number of iterations:	100
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Loading

1 Distributed Load present

Distributed Load 1	
Distribution:	Constant
Magnitude [kPa]:	50
Orientation:	Vertical

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	20 kN/m3
Cohesion	3 kPa
Phi	19.6 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	0.903987
Center:	27.649, 61.585
Radius:	36.661
Left Slip Surface Endpoint:	30.000, 25.000
Right Slip Surface Endpoint:	52.892, 35.000
Resisting Moment:	16847 kN-m
Driving Moment:	18636.3 kN-m
Total Slice Area:	51.1476 m2
Surface Horizontal Width:	22.8915 m
Surface Average Height:	2.23435 m

Method: spencer

FS	0.903192
Center:	27.649, 61.585
Radius:	36.661
Left Slip Surface Endpoint:	30.000, 25.000
Right Slip Surface Endpoint:	52.892, 35.000
Resisting Moment:	16832.1 kN-m
Driving Moment:	18636.3 kN-m
Resisting Horizontal Force:	409.019 kN
Driving Horizontal Force:	452.859 kN
Total Slice Area:	51.1476 m2
Surface Horizontal Width:	22.8915 m
Surface Average Height:	2.23435 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	9792
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	9437
Number of Invalid Surfaces:	355

Error Codes

Error Code -108 reported for 275 surfaces
Error Code -111 reported for 80 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.903987

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.915661	3.54766	4.39578	Material 1	3	19.6	4.7024	4.25091	3.51295	0	3.51295	3.87443	3.87443
2	0.915661	10.431	5.8327	Material 1	3	19.6	7.50396	6.78348	10.6252	0	10.6252	11.3918	11.3918
3	0.915661	16.8888	7.27332	Material 1	3	19.6	10.0773	9.10977	17.1583	0	17.1583	18.4444	18.4444
4	0.915661	22.9173	8.71858	Material 1	3	19.6	12.4267	11.2335	23.1225	0	23.1225	25.0282	25.0282
5	0.915661	28.5119	10.1695	Material 1	3	19.6	14.5556	13.1581	28.5271	0	28.5271	31.1381	31.1381
6	0.915661	33.6671	11.627	Material 1	3	19.6	16.4671	14.886	33.3799	0	33.3799	36.7682	36.7682
7	0.915661	38.3764	13.0921	Material 1	3	19.6	18.1638	16.4198	37.6871	0	37.6871	41.9113	41.9113
8	0.915661	42.6322	14.5661	Material 1	3	19.6	19.6474	17.761	41.4537	0	41.4537	46.5591	46.5591
9	0.915661	46.4258	16.05	Material 1	3	19.6	20.9198	18.9112	44.6838	0	44.6838	50.7022	50.7022
10	0.915661	49.7472	17.545	Material 1	3	19.6	21.9817	19.8712	47.3797	0	47.3797	54.3295	54.3295
11	0.915661	52.5852	19.0525	Material 1	3	19.6	22.8338	20.6415	49.5432	0	49.5432	57.4289	57.4289
12	0.915661	54.9269	20.5738	Material 1	3	19.6	23.4764	21.2224	51.1744	0	51.1744	59.9864	59.9864
13	0.915661	56.7579	22.1104	Material 1	3	19.6	23.909	21.6134	52.2725	0	52.2725	61.986	61.986
14	0.915661	58.0617	23.664	Material 1	3	19.6	24.1307	21.8138	52.8354	0	52.8354	63.4099	63.4099
15	0.915661	58.82	25.2363	Material 1	3	19.6	24.1404	21.8226	52.8598	0	52.8598	64.2381	64.2381
16	0.915661	59.0119	26.8292	Material 1	3	19.6	23.9361	21.6379	52.3414	0	52.3414	64.4477	64.4477
17	0.915661	58.6137	28.4449	Material 1	3	19.6	23.5157	21.2579	51.2742	0	51.2742	64.0129	64.0129
18	0.915661	57.5988	30.0856	Material 1	3	19.6	22.8764	20.68	49.6512	0	49.6512	62.9045	62.9045
19	0.915661	55.9365	31.7541	Material 1	3	19.6	22.0148	19.9011	47.4637	0	47.4637	61.0891	61.0891
20	0.915661	53.592	33.4532	Material 1	3	19.6	20.9268	18.9176	44.702	0	44.702	58.5286	58.5286
21	0.915661	50.5251	35.1864	Material 1	3	19.6	19.6082	17.7256	41.3542	0	41.3542	55.1794	55.1794
22	0.915661	46.5847	36.9574	Material 1	3	19.6	20.4227	18.4619	43.4221	0	43.4221	58.7879	58.7879
23	0.915661	36.5127	38.7707	Material 1	3	19.6	29.4151	26.5909	66.2509	0	66.2509	89.8765	89.8765
24	0.915661	22.5844	40.6313	Material 1	3	19.6	24.4617	22.1131	53.6758	0	53.6758	74.6652	74.6652
25	0.915661	7.69508	42.5454	Material 1	3	19.6	19.3346	17.4782	40.6594	0	40.6594	58.4045	58.4045

Global Minimum Query (spencer) - Safety Factor: 0.903192

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.915661	3.54766	4.39578	Material 1	3	19.6	5.61433	5.07082	5.81555	0	5.81555	6.24713	6.24713
2	0.915661	10.431	5.8327	Material 1	3	19.6	8.79401	7.94268	13.8806	0	13.8806	14.779	14.779
3	0.915661	16.8888	7.27332	Material 1	3	19.6	11.6265	10.501	21.0653	0	21.0653	22.5491	22.5491
4	0.915661	22.9173	8.71858	Material 1	3	19.6	14.1319	12.7638	27.4201	0	27.4201	29.5872	29.5872
5	0.915661	28.5119	10.1695	Material 1	3	19.6	16.328	14.7473	32.9903	0	32.9903	35.9192	35.9192
6	0.915661	33.6671	11.627	Material 1	3	19.6	18.2307	16.4658	37.8163	0	37.8163	41.5674	41.5674
7	0.915661	38.3764	13.0921	Material 1	3	19.6	19.8538	17.9318	41.9334	0	41.9334	46.5507	46.5507
8	0.915661	42.6322	14.5661	Material 1	3	19.6	21.2102	19.1569	45.3739	0	45.3739	50.8854	50.8854
9	0.915661	46.4258	16.05	Material 1	3	19.6	22.3111	20.1512	48.1661	0	48.1661	54.5848	54.5848
10	0.915661	49.7472	17.545	Material 1	3	19.6	23.1664	20.9237	50.3358	0	50.3358	57.6601	57.6601
11	0.915661	52.5852	19.0525	Material 1	3	19.6	23.7853	21.4827	51.9055	0	51.9055	60.1198	60.1198
12	0.915661	54.9269	20.5738	Material 1	3	19.6	24.1758	21.8354	52.8961	0	52.8961	61.9706	61.9706
13	0.915661	56.7579	22.1104	Material 1	3	19.6	24.3453	21.9885	53.3258	0	53.3258	63.2166	63.2166
14	0.915661	58.0617	23.664	Material 1	3	19.6	24.3001	21.9477	53.2115	0	53.2115	63.8603	63.8603
15	0.915661	58.82	25.2363	Material 1	3	19.6	24.0464	21.7185	52.5677	0	52.5677	63.9017	63.9017
16	0.915661	59.0119	26.8292	Material 1	3	19.6	23.5892	21.3056	51.4082	0	51.4082	63.3391	63.3391
17	0.915661	58.6137	28.4449	Material 1	3	19.6	22.9336	20.7134	49.745	0	49.745	62.1684	62.1684
18	0.915661	57.5988	30.0856	Material 1	3	19.6	22.0837	19.9458	47.5893	0	47.5893	60.3833	60.3833
19	0.915661	55.9365	31.7541	Material 1	3	19.6	21.0435	19.0063	44.9509	0	44.9509	57.9751	57.9751
20	0.915661	53.592	33.4532	Material 1	3	19.6	19.8167	17.8983	41.8393	0	41.8393	54.9325	54.9325
21	0.915661	50.5251	35.1864	Material 1	3	19.6	18.4069	16.625	38.2634	0	38.2634	51.2415	51.2415
22	0.915661	46.5847	36.9574	Material 1	3	19.6	18.9212	17.0895	39.5678	0	39.5678	53.804	53.804
23	0.915661	36.5127	38.7707	Material 1	3	19.6	26.5746	24.002	58.9805	0	58.9805	80.3247	80.3247
24	0.915661	22.5844	40.6313	Material 1	3	19.6	21.9609	19.8349	47.2778	0	47.2778	66.1214	66.1214
25	0.915661	7.69508	42.5454	Material 1	3	19.6	17.3203	15.6436	35.5073	0	35.5073	51.4038	51.4038

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.903987

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30.0004	25.0002	0	0	0
2	30.916	25.0706	4.05839	0	0
3	31.8317	25.1641	9.9354	0	0
4	32.7473	25.281	17.1573	0	0
5	33.663	25.4214	25.2887	0	0
6	34.5787	25.5856	33.9307	0	0
7	35.4943	25.774	42.7195	0	0
8	36.41	25.987	51.3254	0	0
9	37.3256	26.2249	59.4519	0	0
10	38.2413	26.4884	66.8358	0	0
11	39.157	26.7778	73.2467	0	0
12	40.0726	27.0941	78.4873	0	0
13	40.9883	27.4378	82.3946	0	0
14	41.9039	27.8098	84.8408	0	0
15	42.8196	28.211	85.7348	0	0
16	43.7353	28.6426	85.0249	0	0
17	44.6509	29.1057	82.7012	0	0
18	45.5666	29.6018	78.7998	0	0
19	46.4822	30.1322	73.4071	0	0
20	47.3979	30.699	66.6659	0	0
21	48.3136	31.304	58.7831	0	0
22	49.2292	31.9496	50.0386	0	0
23	50.1449	32.6385	38.8234	0	0
24	51.0606	33.3739	17.0334	0	0
25	51.9762	34.1596	-2.74094	0	0
26	52.8919	35	0	0	0

Global Minimum Query (spencer) - Safety Factor: 0.903192

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30.0004	25.0002	0	0	0
2	30.916	25.0706	4.73131	2.17311	24.6695
3	31.8317	25.1641	11.485	5.27511	24.6695
4	32.7473	25.281	19.6688	9.03395	24.6695
5	33.663	25.4214	28.7581	13.2087	24.6695
6	34.5787	25.5856	38.2899	17.5867	24.6695
7	35.4943	25.774	47.8576	21.9811	24.6694
8	36.41	25.987	57.1067	26.2293	24.6695
9	37.3256	26.2249	65.7315	30.1907	24.6695
10	38.2413	26.4884	73.472	33.7459	24.6695
11	39.157	26.7778	80.1118	36.7956	24.6695
12	40.0726	27.0941	85.4765	39.2597	24.6695
13	40.9883	27.4378	89.4325	41.0767	24.6695
14	41.9039	27.8098	91.8863	42.2037	24.6695
15	42.8196	28.211	92.7846	42.6163	24.6695
16	43.7353	28.6426	92.1147	42.3086	24.6695
17	44.6509	29.1057	89.9056	41.294	24.6695
18	45.5666	29.6018	86.2296	39.6056	24.6695
19	46.4822	30.1322	81.2048	37.2976	24.6695
20	47.3979	30.699	74.9983	34.447	24.6695
21	48.3136	31.304	67.8308	31.1549	24.6695
22	49.2292	31.9496	59.9818	27.5498	24.6694
23	50.1449	32.6385	50.047	22.9868	24.6695
24	51.0606	33.3739	31.003	14.2398	24.6695
25	51.9762	34.1596	13.9656	6.41444	24.6695
26	52.8919	35	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	70, 20 70, 35 50, 35 30, 25 20, 25 20, 20

Scenario-based Entities

Type	Coordinates (x,y)	Master Scenario
Distributed Load	50, 35 70, 35	Constant DistributionOrientation: VerticalMagnitude: 50 kN/m2Creates Excess Pore Pressure: No